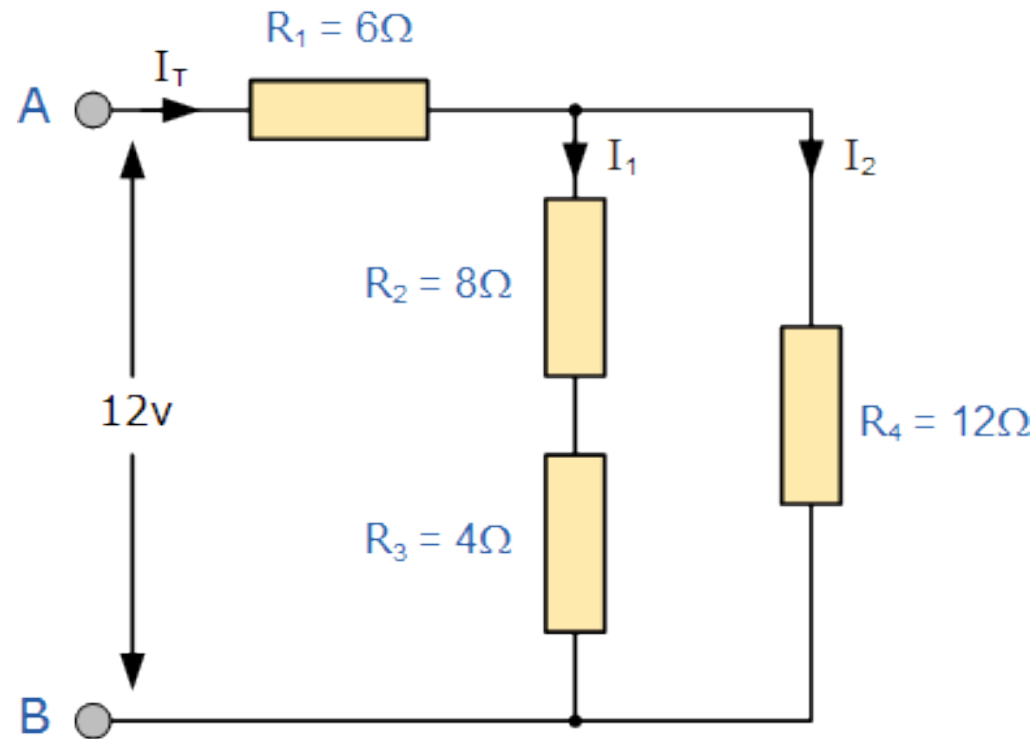


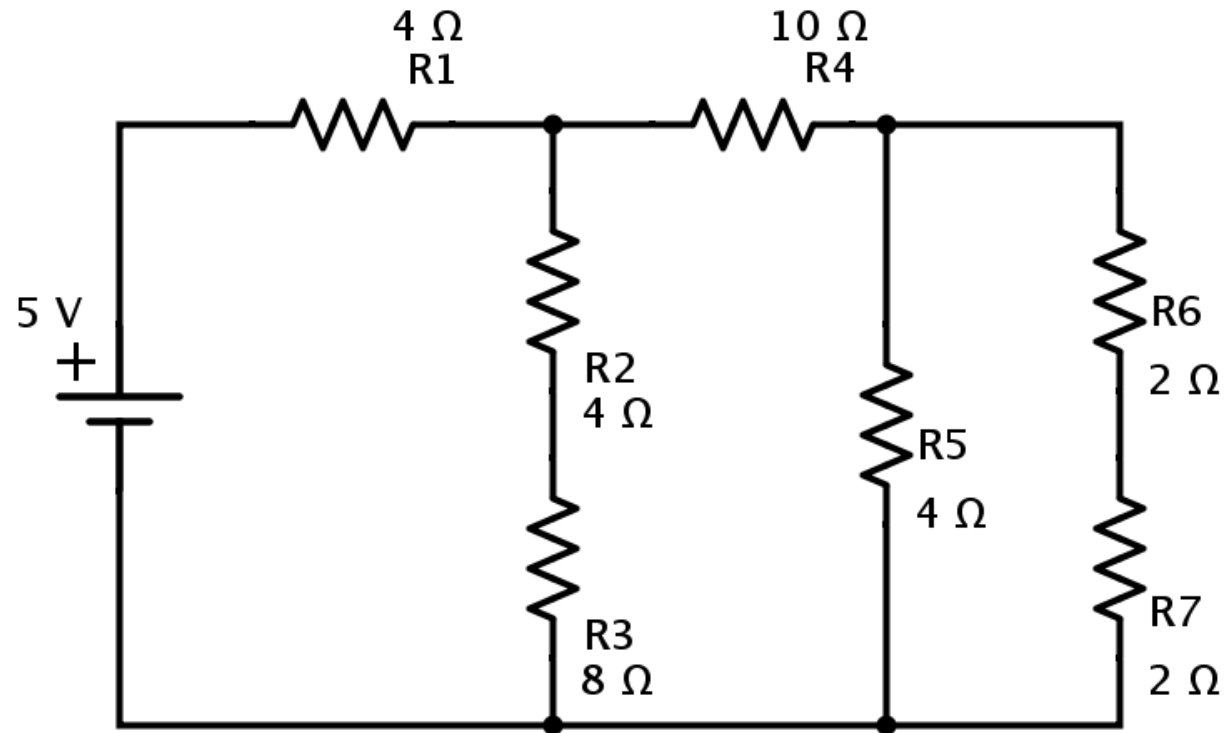
Series and Parallel Resistor Comb. Contd.

- A little complex example



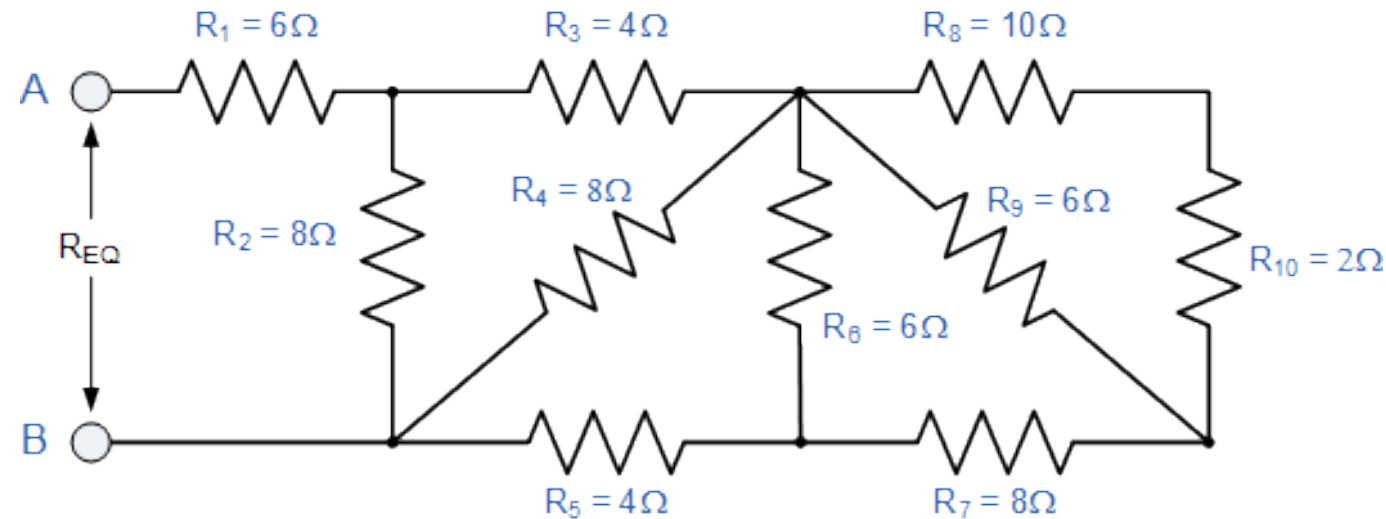
Series and Parallel Resistor Comb. Contd.

○ Task for you



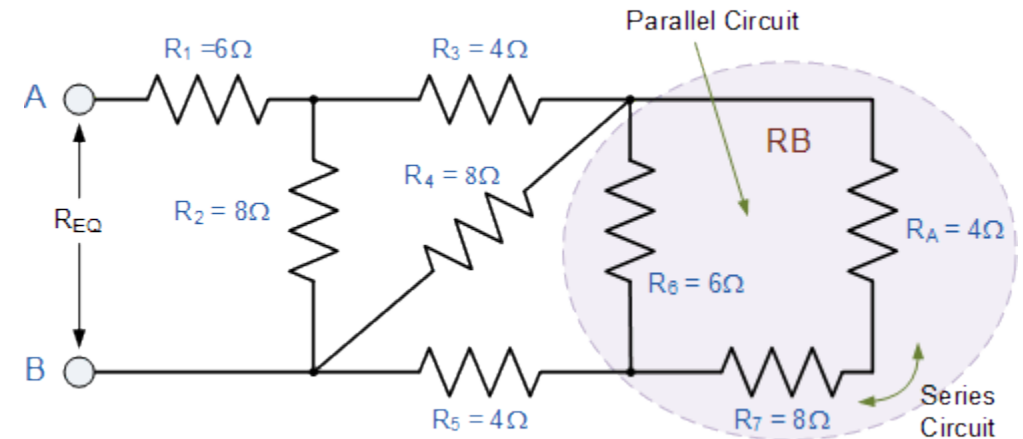
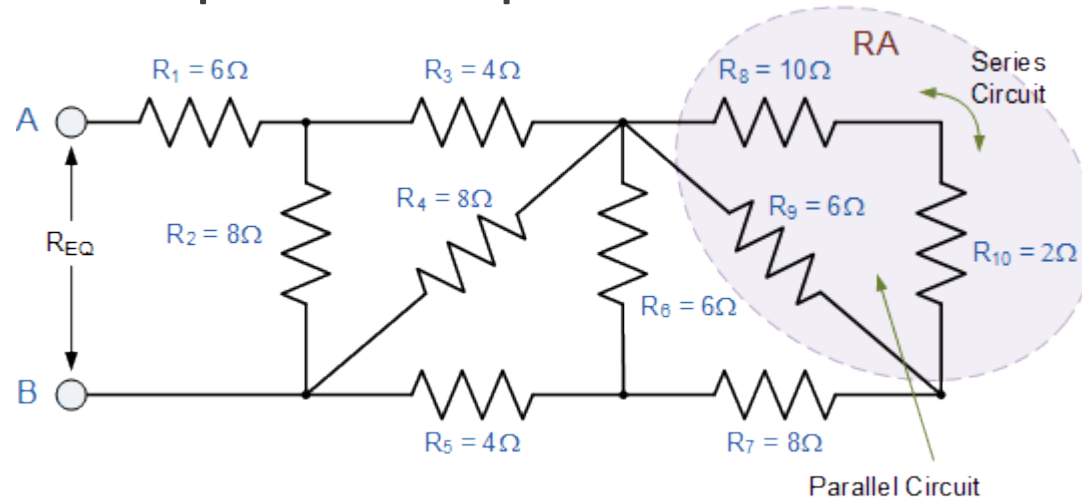
Series and Parallel Resistor Comb. Contd.

- A complex example



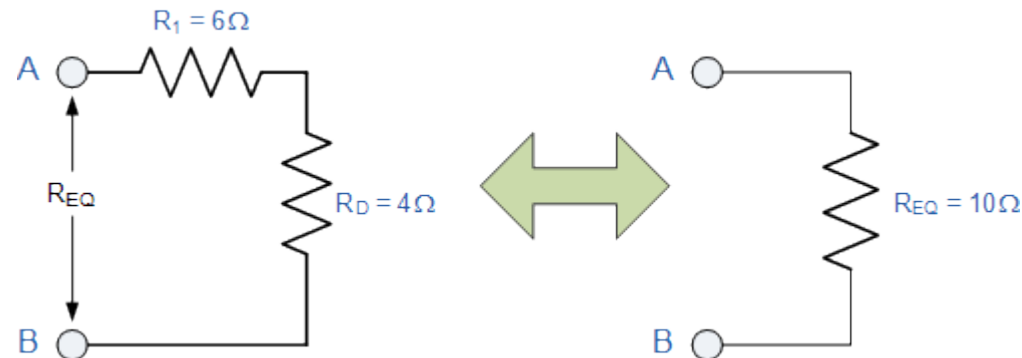
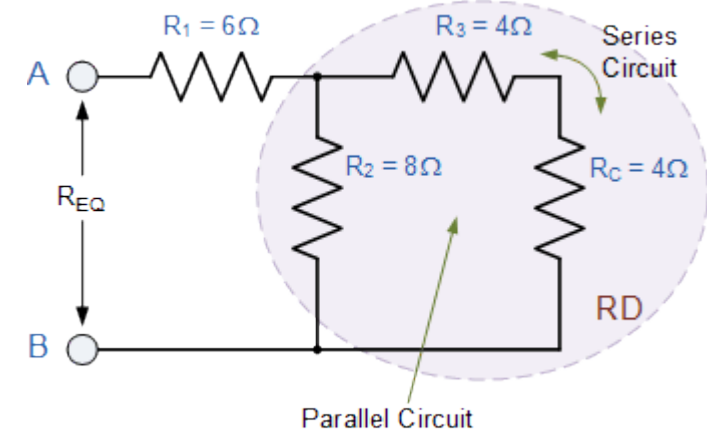
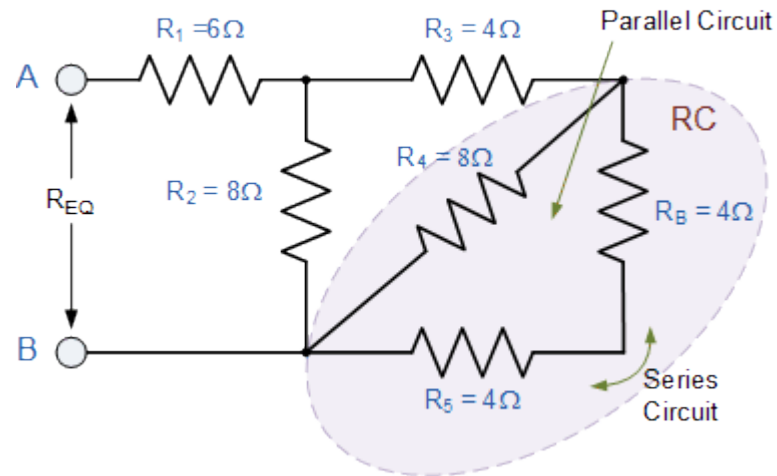
Series and Parallel Resistor Comb. Contd.

○ A complex example



Series and Parallel Resistor Comb. Contd.

○ A complex example



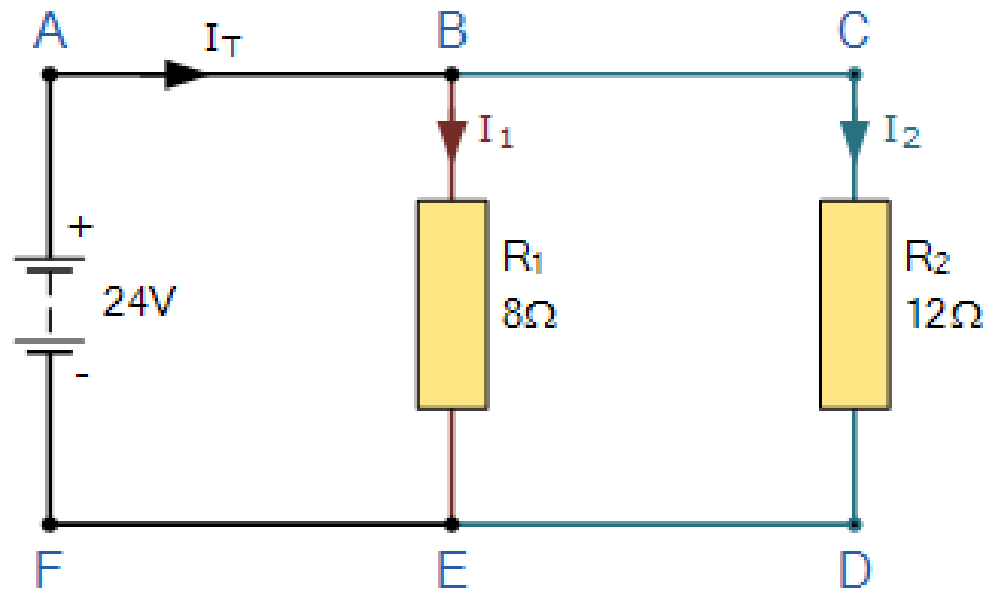
Kirchhoff's Current Law (KCL)

- Kirchhoff's Current Law is one of the fundamental law used for circuit analysis
- It states that the total current entering a circuits node is exactly equal to the total current leaving the same node
- Mathematically we can write it as

$$\sum I_{IN} = \sum I_{OUT}$$

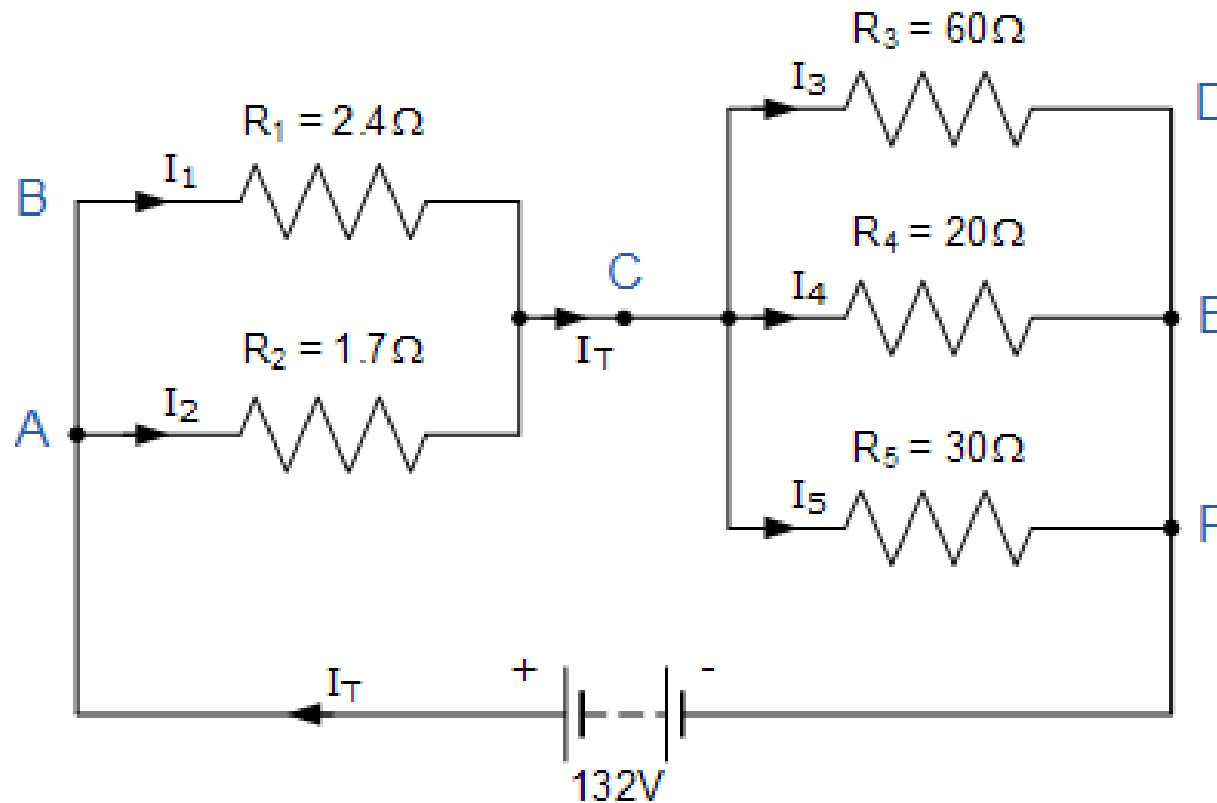
Kirchhoff's Current Law (KCL) contd.

- lets take a simple example



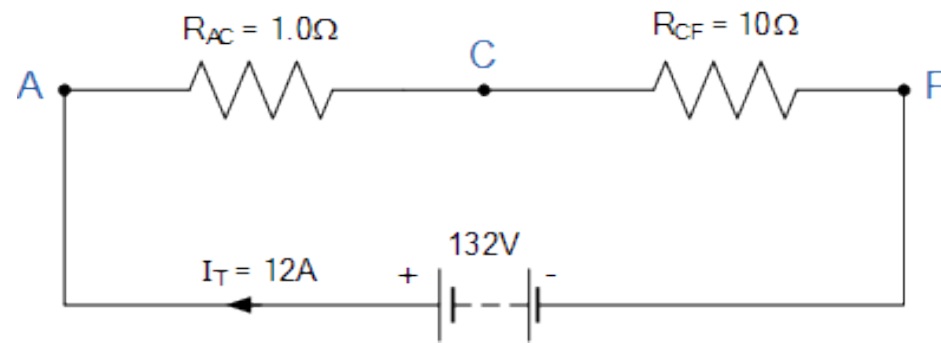
Kirchhoff's Current Law (KCL) contd.

- lets take a complex example



Kirchhoff's Current Law (KCL) contd.

- Equivalent circuit will be



$$V_{AC} = I_T \times R_{AC} = 12 \times 1 = 12 \text{ Volts}$$

$$V_{CF} = I_T \times R_{CF} = 12 \times 10 = 120 \text{ Volts}$$

$$I_1 = \frac{V_{AC}}{R_1} = \frac{12}{2.4} = 5 \text{ Amps}$$

$$I_2 = \frac{V_{AC}}{R_2} = \frac{12}{1.7} = 7 \text{ Amps}$$

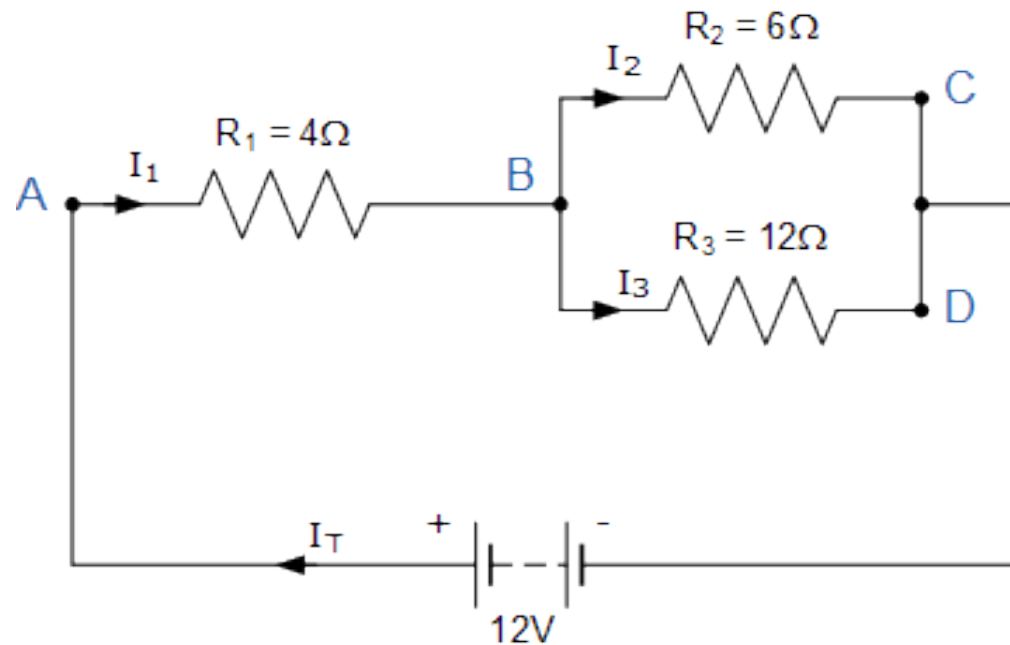
$$I_3 = \frac{V_{CF}}{R_3} = \frac{120}{60} = 2 \text{ Amps}$$

$$I_4 = \frac{V_{CF}}{R_4} = \frac{120}{20} = 6 \text{ Amps}$$

$$I_5 = \frac{V_{CF}}{R_5} = \frac{120}{30} = 4 \text{ Amps}$$

Kirchhoff's Current Law (KCL) contd.

- Lets take another simple example

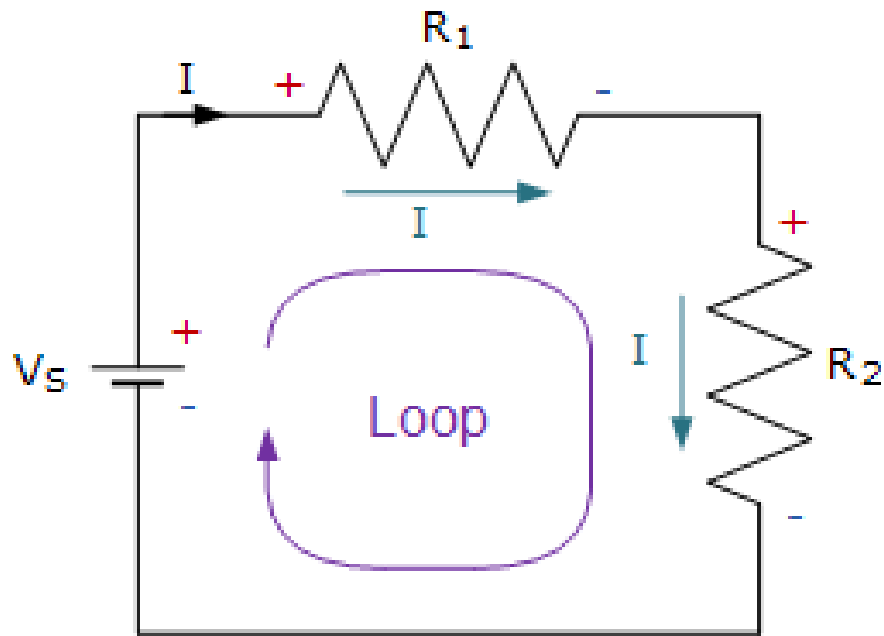


Kirchhoff's Voltage Law (KVL)

- Kirchhoff's Voltage Law is the second of his fundamental laws we can use for circuit analysis
- It states that for a closed loop series path the algebraic sum of all the voltages around any closed loop in a circuit is equal to zero
- Mathematically we can write it as

$$\sum V = 0 \text{ for a closed loop}$$

Kirchhoff's Voltage Law (KVL) contd.



$$V_S + (-IR_1) + (-IR_2) = 0$$

$$\therefore V_S = IR_1 + IR_2$$

$$V_S = I(R_1 + R_2)$$

$$V_S = IR_T$$

$$\text{Where: } R_T = R_1 + R_2$$

Kirchhoff's Voltage Law (KVL) contd.

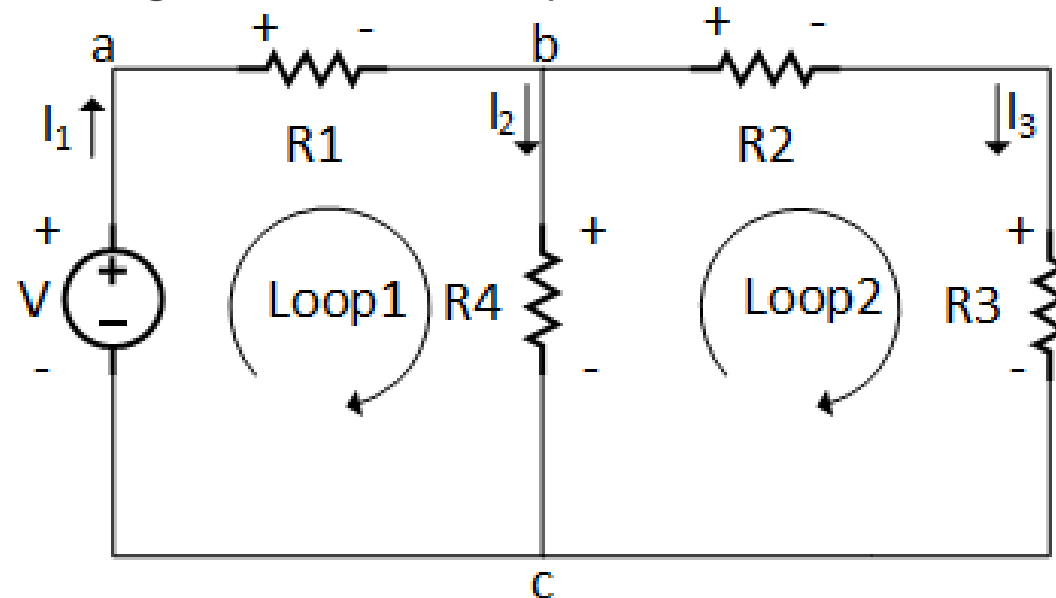
Three resistor of values: 10 ohms, 20 ohms and 30 ohms, respectively are connected in series across a 12 volt battery supply. Calculate: a) the total resistance, b) the circuit current, c) the current through each resistor, d) the voltage drop across each resistor e) verify that Kirchhoff's voltage law, KVL holds true.

Kirchhoff's Voltage Law (KVL) contd.

Kirchhoff's Voltage Law (KVL) contd.

- Another example with two loops where we have values of resistors and voltage source as follows:

$R_1=5\Omega$, $R_2=10\Omega$, $R_3=5\Omega$ and $R_4=10\Omega$ and $V=20V$



Kirchhoff's Voltage Law (KVL) contd.

Assign Current Names:

$$V = (R_1 + R_4)I_2 + R_1I_3$$

$$0 = -R_4I_2 + (R_3 + R_2)I_3$$

at node a: $I_1 = I_1$

Let's put the values of resistors and the voltage source, and see what happen

at node b: $I_1 = I_2 + I_3$

$$20 = (5 + 10)I_2 + 5I_3 \text{--- (1)}$$

at node c: $I_2 + I_3 = I_1$

$$0 = -10I_2 + (5 + 10)I_3 \text{--- (2)}$$

Write Loop Equations:

Now multiply the equation (1) by -3 and add with the equation (2)

The next step is to write equations for each loop.

$$0 = -10I_2 + (15)I_3 \text{--- (2)}$$

For loop1: $V = I_1R_1 + I_2R_4$

$$-60 = -45I_2 - 15I_3 \text{--- (1)}$$

For loop2: $0 = I_3R_2 + I_3R_3 - I_2R_4$

Kirchhoff's Voltage Law (KVL) contd.

Solve The Equations:

Adding the above two equation will give us the equation (3)

$$-60 = -55I_2 + 0 - (3)$$

OR

$$I_2 = -\frac{60}{-55} = 1.09Amp$$

Now putting the current I_2 in the equation (2) to get current I_3

$$0 = -10(1.09) + 15I_3 \quad \text{And}$$

OR

$$I_3 = 10 \frac{(1.09)}{15} = 0.72Amp$$

By applying KCL to the node b, the following equation can be obtained

$$I_1 = I_2 + I_3$$

Now putting the values of the current will give us the value of I_1

$$I_1 = 1.09 + 0.72 = 1.81Amp$$

Now, it is easy to find any parameter of the circuit, just use the ohm's law, as shown below

$$V_3 = I_3 R_3 = 0.72(5) = 3.6Volts$$

$$V_2 = I_3 R_2 = 0.72(10) = 7.2Volts$$

$$V_4 = I_2 R_4 = 1.09(10) = 10.9Volts$$

$$V_1 = I_1 R_1 = 1.81(5) = 9.05Volts$$